

LAB EXPERIMENTS

Q14) Write a C program for one-time pad version of the Vigenère cipher. In this scheme, the key is a stream of random numbers between 0 and 26. For example, if the key is 3 19 5 . . . , then the first letter of plaintext is encrypted with a shift of 3 letters, the second with a shift of 19 letters, the third with a shift of 5 letters, and so on.

Aim:

To write a C program to implement the **One-Time Pad (OTP)** version of the **Vigenère Cipher**, where each plaintext letter is encrypted using a unique random key value (0–25) that is used only once.

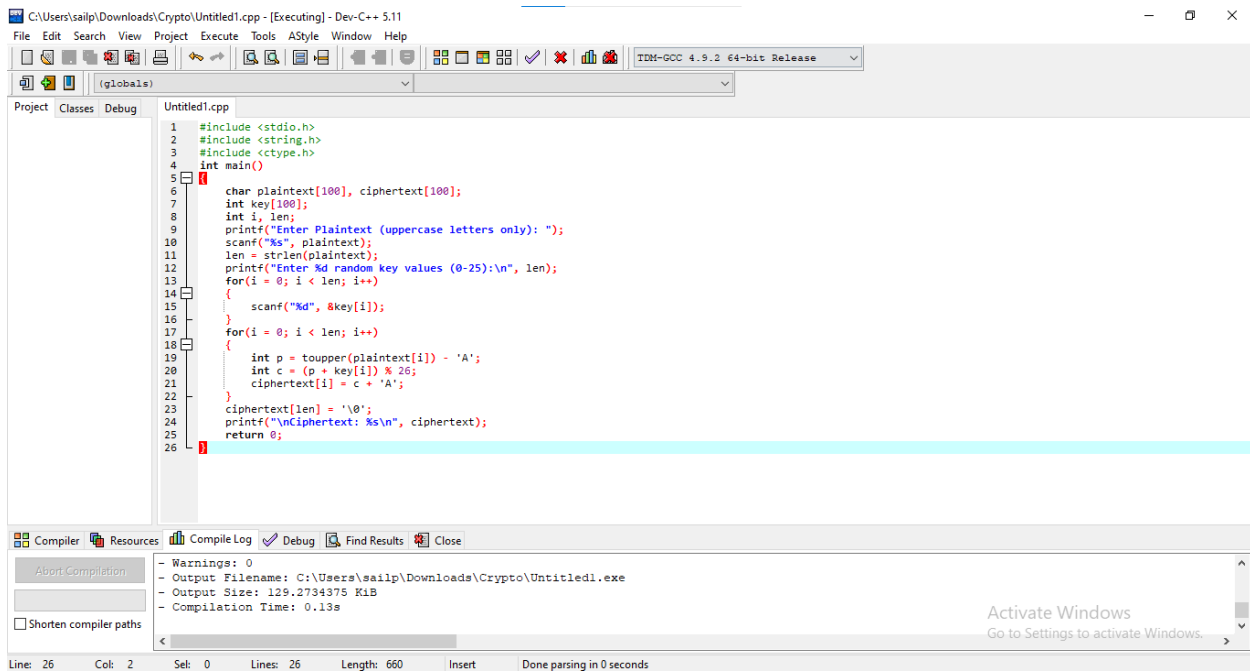
Algorithm:

1. Start the program.
2. Read the plaintext message from the user.
3. Read the random key values (0–25), one for each plaintext character.
4. Convert all plaintext letters to uppercase.
5. Convert each plaintext letter into its numerical equivalent (A = 0, B = 1, ..., Z = 25).
6. Encrypt each letter using the formula:
$$C = (P + K) \bmod 26$$
where **P** is the plaintext value and **K** is the corresponding random key.
7. Convert the encrypted numerical value back into its alphabetic character.
8. Repeat the process for every character using its corresponding key value.
9. Display the ciphertext.
10. Stop the program.

Result:

Thus, the C program to implement the **One-Time Pad (OTP)** version of the **Vigenère Cipher** was successfully executed. The plaintext was encrypted using a unique random key for each character, producing the corresponding ciphertext.

Code:



The screenshot shows a C++ IDE with the following code in `Untitled1.cpp`:

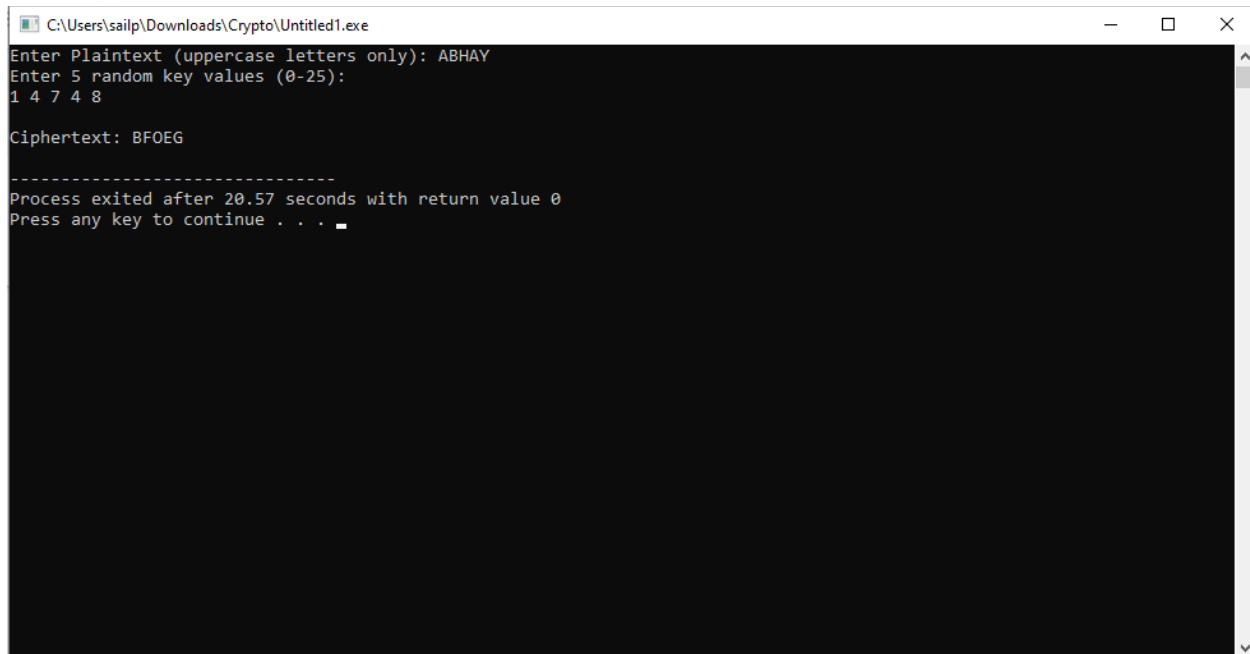
```
1 #include <stdio.h>
2 #include <string.h>
3 #include <ctype.h>
4 int main()
5 {
6     char plaintext[100], ciphertext[100];
7     int key[100];
8     int i, len;
9     printf("Enter Plaintext (uppercase letters only): ");
10    scanf("%s", plaintext);
11    len = strlen(plaintext);
12    printf("Enter %d random key values (0-25):\n", len);
13    for(i = 0; i < len; i++)
14    {
15        scanf("%d", &key[i]);
16    }
17    for(i = 0; i < len; i++)
18    {
19        int p = toupper(plaintext[i]) - 'A';
20        int c = (p + key[i]) % 26;
21        ciphertext[i] = c + 'A';
22    }
23    ciphertext[len] = '\0';
24    printf("\nCiphertext: %s\n", ciphertext);
25    return 0;
26 }
```

The IDE interface includes a menu bar (File, Edit, Search, View, Project, Execute, Tools, AStyle, Window, Help), a toolbar, a project explorer showing `(globals)`, and a compiler output window at the bottom. The compiler output shows:

```
- Warnings: 0
- Output Filename: C:\Users\sailp\Downloads\Crypto\Untitled1.exe
- Output Size: 129,273,4375 KiB
- Compilation Time: 0.13s
```

The status bar at the bottom indicates: Line: 26, Col: 2, Sel: 0, Lines: 26, Length: 660, Insert, Done parsing in 0 seconds.

Output:



The screenshot shows a Windows command prompt window titled `C:\Users\sailp\Downloads\Crypto\Untitled1.exe`. The output of the program is as follows:

```
Enter Plaintext (uppercase letters only): ABHAY
Enter 5 random key values (0-25):
1 4 7 4 8

Ciphertext: BFOEG

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Process exited after 20.57 seconds with return value 0
Press any key to continue . . .
```